



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

15-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C	A,C	C	A,B,D	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.16 to 0.17	1.00	1.51	0.50	1.41	4.50	0.50	0.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C	A,C,D	A,C,D	A,B	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	A	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	23.30	1.50	7.00	2.00	78.00	49.25	6.90 to 6.92	0.57

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B	A,B	A,C	A,C	A,B,C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	5.00	5.00	4.00	0.00	0.00	5.00	0.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C)

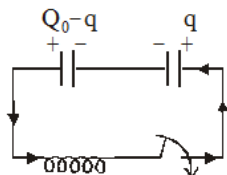
$$-L \frac{di}{dt} - \frac{q}{c} + \frac{(Q_0 - q)}{c} = 0$$

$$\frac{-Ld^2q}{dt^2} - \frac{2q}{c} + \frac{Q_0}{c} = 0$$

$$\frac{d^2q}{dt^2} = -\frac{2}{LC} \left[q - \frac{Q_0}{2} \right]$$

$$\text{frequency} = \frac{1}{\pi\sqrt{2LC}}$$

$$\text{max current} = \frac{Q_0}{2} \omega = \frac{d_0}{2} \sqrt{\frac{2}{LC}} = \frac{Q_0}{2LC}$$



2. Ans (A,C)

Current in secondary circuit

$$i = \frac{3}{R + 2R} = \frac{3}{3R} = \frac{1}{R}$$

$$\text{current through } 3R = \frac{2 \times (1/R)}{3} = \frac{2}{3R}$$

$$\text{P.d. across } xy = \frac{2}{3R} \times 3R = 2 \text{ volt}$$

$$\text{P.d. across A and C} = \frac{25}{100} \left[\frac{10}{r+8} \right] \times 8$$

$$2 = \frac{1}{4} \left[\frac{10}{r+8} \right] \times 8 \quad 3r + 32 = 40 \quad r = 2\Omega$$

3. Ans (A,C)

$$\text{So } t = \frac{2\ell + \pi R}{V_0}$$

4. Ans (C)

$$Z_{RL} = \sqrt{R^2 + \omega^2 L^2}$$

$$Z_{RLC} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\cos \phi_{RL} = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}$$

$$\cos \phi_{RLC} = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

According to question

$$(\omega L)^2 = \left(\omega L - \frac{1}{C\omega}\right)^2 \Rightarrow C = \frac{1}{2\omega^2 L}$$

5. Ans (A,B,D)

From 2nd graph, when $\theta = \frac{\pi}{2}$,

$\phi = 4\pi = 2\lambda = \text{distance between } S_1 \text{ \& } S_2$

At $\theta = \frac{\pi}{6}$, $\phi = d \sin \theta = 2\lambda \frac{\ell}{2} = \lambda$.

6. Ans (A,B,C)

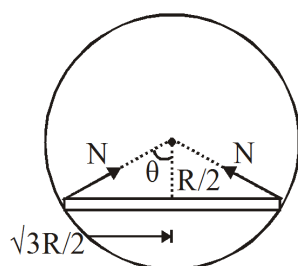
$$\frac{1}{2} \left[\frac{M(\sqrt{3}R)^2}{12} + M \left[\frac{R}{2} \right]^2 \right] \omega^2 = mg \frac{R}{2}$$

$$\Rightarrow \frac{1}{2} \left[\frac{MR^2}{4} + \frac{MR^2}{4} \right] \omega^2 = mg \frac{R}{2}$$

$$\Rightarrow \frac{MR^2}{2} \omega^2 = MgR$$

$$\Rightarrow \omega = \sqrt{\frac{2g}{R}}$$

$$V_{cm} = \omega \frac{R}{2} = \sqrt{\frac{2g}{R}} \times \frac{R^2}{4} = \sqrt{\frac{9R}{2}}$$



$$2N \cos \theta = M \omega^2 \frac{R}{2}$$

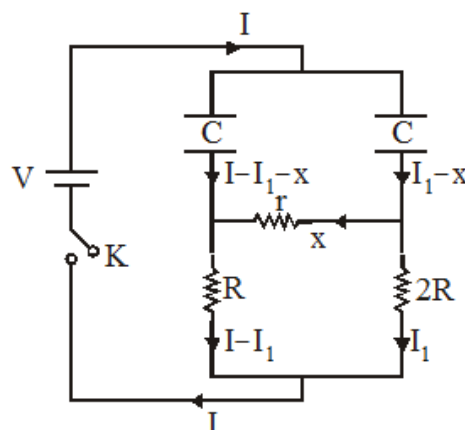
$$2N \times \frac{R/2}{R} = M \times \frac{2g}{R} \times \frac{R}{2}$$

$$N = Mg$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)



Our target is $\int_0^\infty x dt$

$\int_0^\infty I dt = 2 CV$ Now Let $\int_0^\infty x dt = q$ and

$$\int_0^\infty I_1 dt = q_1$$

Now applying loop rule $2I_1 R = rx + R(I - I_1)$

$$\Rightarrow 3I_1 = \frac{r}{R} x + I$$

$$3 \int_0^\infty I_1 dt = \frac{r}{R} \int_0^\infty x dt + \int_0^\infty I dt$$

$$\Rightarrow 3 q_1 = \frac{r}{R} q + 2CV \quad (1)$$

$$\text{Also } \int_0^\infty (I_1 + x) dt = CV$$

$$\therefore q_1 + q = CV \quad (2)$$

$$\text{Solving (1) and (2) } q = \frac{CV}{3 + r/R}$$

8. Ans (C)

For (A) : $2A = 0.06 \Rightarrow$ Amplitude of constituent wave $A = 0.03$ m

For (B) : Position of node when $y = 0$

$$\Rightarrow \sin(2\pi x) = 0 \Rightarrow 2\pi x = n\pi \text{ where } n=1,2,3\dots$$

$$\Rightarrow x = \frac{n}{2} = \frac{1}{2} = 0.5 \text{ m}$$

For (C) : Position of antinode

$$0.06 = 0.06 \sin(2\pi x)$$

$$\Rightarrow 2\pi x = (2n-1)\frac{\pi}{2} \text{ where } n=1,2,3\dots$$

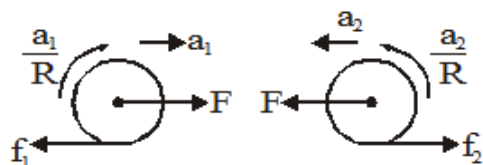
$$\Rightarrow x = \frac{1}{4} = 0.25 \text{ m}$$

For (D) : Amplitude at $x = \frac{1}{12}$ m

$$\Rightarrow A = 0.06 \sin\left(2\pi \times \frac{1}{12}\right) = 0.06 \sin \frac{\pi}{6}$$

$$= 0.06 \times \frac{1}{2} = 0.03 \text{ m}$$

9. Ans (A)



Let v_1 = Velocity of A and v_2 = velocity of B just before they strike by energy conservation

$$0 = \frac{k(2\theta)(-\theta)}{2R} + \frac{1}{2}(2m)v_1^2 \left[1 + \frac{2}{5}\right]$$

$$+ \frac{1}{2}(m)v_2^2 \left[1 + \frac{2}{3}\right]$$

$$\Rightarrow \frac{7}{5}mv_1^2 + \frac{5}{6}mv_2^2 = \frac{kQ^2}{R}$$

By taking torque about contact points

$$FR = \left[\frac{7}{5}(2m)R^2\right] \frac{a_1}{R} \text{ \& } FR = \left[\frac{5}{3}(m)R^2\right] \frac{a_2}{R}$$

$$\Rightarrow 42a_1 = 25a_2 \Rightarrow 42v_1 = 25v_2$$

$$\Rightarrow \frac{7}{5}mv_1^2 + \frac{5}{6}\left(\frac{42}{25}\right)^2 v_1^2 = \frac{kQ^2}{R}$$

$$\Rightarrow \frac{469}{125}mv_1^2 = \frac{kQ^2}{R} \Rightarrow mv_1^2 = \frac{125kQ^2}{469R}$$

Therefore

$$\text{For (A) KE of system} = \frac{kQ^2}{R}$$

$$\text{For (B) KE of A} = \frac{7}{5}mv_1^2 = \frac{25kQ^2}{67R}$$

$$\text{For (C) KE of B} = \frac{5}{6}mv_2^2 = \frac{42kQ^2}{67R}$$

For (D) Rotational KE of system

$$= \frac{2}{5}mv_1^2 + \frac{1}{3}mv_2^2 = \frac{838kQ^2}{2345R}$$

10. Ans (D)

For A :

$$B_{P_1} = 4 \times \frac{\mu_0 I}{4\pi(L/2)} \left[\sin \frac{\pi}{4} + \sin \frac{\pi}{4} \right] = \frac{2\sqrt{2}\mu_0 I}{\pi L}$$

For B :

$$B_{P_2} = \frac{2kM}{r^3} = \frac{2\left(\frac{\mu_0}{4\pi}\right)(IL^2)}{a^3} = \frac{\mu_0 IL^2}{2\pi a^3}$$

For C :

$$B_{P_3} = \frac{kM}{r^3} = \frac{\left(\frac{\mu_0}{4\pi}\right)(IL^2)}{a^3} = \frac{\mu_0 IL^2}{4\pi a^3}$$

For D :

$$B_{P_4} = \frac{kM}{r^3} \sqrt{1+3\cos^2\theta} \text{ where } \cos\theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow B_{P_4} = \frac{\left(\frac{\mu_0}{4\pi}\right)(IL^2)}{(a\sqrt{2})^3} \sqrt{1+3\left(\frac{1}{2}\right)} = \frac{\mu_0 \sqrt{5} IL^2}{16\pi a^3}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (0.16 to 0.17)

$$F = \sigma \phi$$

$$= \frac{Q}{a^2} \times \frac{q}{6\epsilon_0}$$

2. Ans (1.00)

$$v^2 = I \Rightarrow (v_0 - Ir)^2 = I$$

3. Ans (1.51)

$$\cos A = \frac{1}{8} \Rightarrow 1 - 2\sin^2 \frac{A}{2} = \frac{1}{8}$$

$$\Rightarrow \sin \frac{A}{2} = \frac{\sqrt{7}}{4}$$

$$\text{now, required } \mu = \operatorname{cosec} \frac{A}{2} = \frac{4}{\sqrt{7}}$$

4. Ans (0.50)

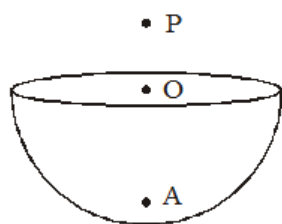
$$Q = dU + W$$

$$= \frac{nR\Delta T}{\gamma - 1} + \frac{nR\Delta T}{1 - \eta}$$

$$= 2RT_0$$

So the value of x is 2.

5. Ans (1.41)



At point p potential is

$$V_P = \frac{-2GM}{R} + \frac{\sqrt{2}GM}{R}$$

$$V_0 = -\frac{GM}{R}, V_A = \frac{-\sqrt{2}GM}{R}$$

$$\text{Now } V_P - V_0 = \frac{GM}{R} + \frac{\sqrt{2}GM}{R} \text{ and } \left[\frac{-GM}{R} + \frac{\sqrt{2}GM}{R} \right]$$

Now applying conservation of energy

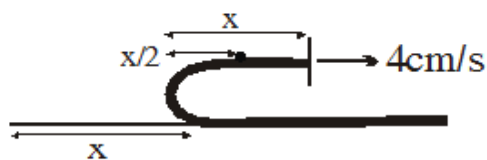
$$\frac{1}{2}mv_0^2 = \left[\frac{-GM}{R} + \frac{\sqrt{2}GM}{R} \right]$$

Again

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + m \left[\frac{-GM}{R} + \frac{\sqrt{2}GM}{R} \right]$$

$$\Rightarrow V = \sqrt{2}V_0$$

6. Ans (4.50)



$$\frac{d(2x)}{dt} = 4\text{m/s}; \frac{2dx}{dt} = 4; \frac{dx}{dt} = 3 \text{ cm/s}$$

$$\frac{d}{dt} \left(x + \frac{x}{2} \right) \Rightarrow \frac{3dx}{2dt} = \frac{3}{2} \times 3 = \frac{9}{2} \text{ cm/sec}$$

7. Ans (0.50)

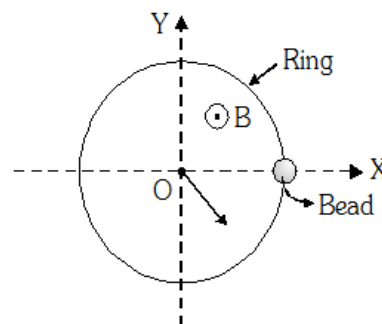
$$E_2 = M \frac{di_1}{dt} = 2ktM$$

$$i_2 = \frac{E_2}{R} = \frac{2ktM}{R}$$

$$\int dq = \int i_2 dt = \frac{2Mk}{R} \int_0^T t dt = \frac{kMT^2}{R}$$

$$\therefore \frac{\lambda}{1-\lambda} = 1 \Rightarrow \lambda = \frac{1}{2} \Rightarrow \frac{1}{\lambda} = 2.$$

8. Ans (0.00)



B depends upon r.

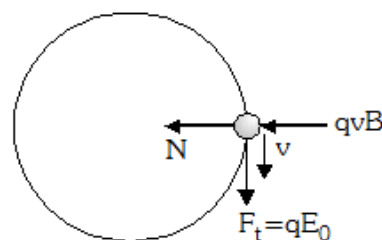
$\therefore$ Flux linked with ring :

$$\phi = \int_0^R B 2\pi r dr = \int_0^R \frac{E_0 t}{r} 2\pi r dr$$

$$\phi = 2\pi E_0 t R$$

$$\frac{d\phi}{dt} = 2\pi E_0 R = (\text{induced electric field}) 2\pi R$$

$\therefore$ Induced electric field = E_0 (tangential)



Circular motion of bead :

$$F_t = qE_0 = \frac{mdV}{dt} \Rightarrow v = \frac{qE_0 t}{m}$$

Centripetal force equation :

$$N + qvB = \frac{mv^2}{R}$$

$$N + q \frac{qE_0 t}{m} \frac{E_0 t}{R} = \frac{m}{R} \frac{q^2 E_0^2 t^2}{m^2}$$

$$N = 0$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (A,C,D)

Difference in refractive indices of DM and DP cause easy detection by ultramicroscope.

Less number of particles gives less (negligible) colligative properties.

4. Ans (A,C,D)

$$\text{pH} = 3 \Rightarrow [\text{H}^+] = 10^{-3} \text{M}$$

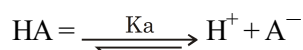
let the degree of dissociation be α

$$\Rightarrow \alpha = \frac{10^{-3}}{0.1} = 0.01 \Rightarrow \% \alpha = 1$$

$$\Rightarrow i = 1 + (y-1)\alpha = 1 + (2-1)(0.01) = 1.01$$

$$\Rightarrow \pi(\text{osmotic pressure}) = (1.01)(0.1)RT = 0.101 RT$$

$$i = \frac{M_{\text{th}}}{M_{\text{ob}}} \Rightarrow M_{\text{th}} = 1.01 \times M_{\text{ob}}$$



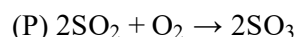
$$K_a = \frac{C\alpha^2}{1-\alpha} \cong 10^{-1} \times 10^{-2} = 10^{-3} \text{M}$$

$$K_b(\text{A}^-) = \frac{K_w}{K_a} = \frac{10^{-14}}{10^{-3}} = 10^{-11} \text{M at } 25^\circ\text{C}$$

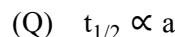
PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (A)



$$\frac{-\frac{d[\text{SO}_2]}{dt}}{2} = \frac{+\frac{d[\text{SO}_3]}{dt}}{2} \Rightarrow \frac{\frac{d(\text{wt. SO}_3)}{dt}}{\frac{d(\text{wt. SO}_2)}{dt}} = 1.2$$



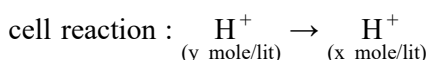
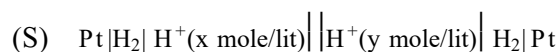
$$\Rightarrow \text{order of reaction} = 0$$

$$(R) t_{1/2} = \frac{0.693}{k} = 1 \text{ min}$$

$$\Rightarrow \frac{0.025}{0.2} = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$$

$$\Rightarrow \frac{1}{8} = \left(\frac{1}{2}\right)^{t/1}$$

$$\Rightarrow t = 3 \text{ min}$$



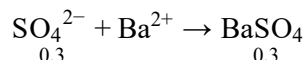
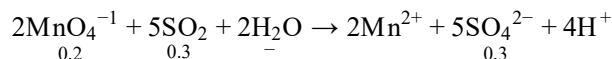
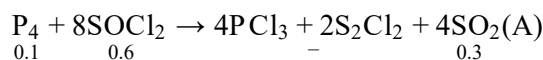
$$E = 0.0177 = \frac{0.059}{1} \log \frac{y}{x}$$

$$\Rightarrow \log \frac{y}{x} = 0.3 \Rightarrow \frac{y}{x} = 2$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (23.30)

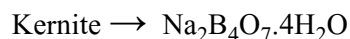
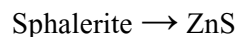
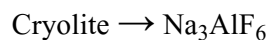
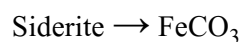
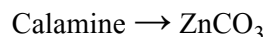
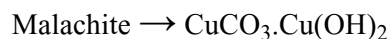


$$\Rightarrow x = 0.3 \times [137 + 32 + 64] = 69.9 \text{g}$$

2. Ans (1.50)

Soluble in KCN $\rightarrow$ CuS, ZnS, Ni(CN)₂, AgCN,
AgCl, CuCN soluble in dil.HCl $\rightarrow$ ZnS, FeS,
MnS

3. Ans (7.00)



$$\Rightarrow 13 \text{ types of elements}$$

4. Ans (2.00)

Benzoic acid + aniline and sulphuric acid

5. Ans (78.00)

$$\text{Molarity of 1}^{\text{st}} \text{ solution} = \frac{0.2 \times M}{\frac{0.2M + 18 \times 0.8}{d_1}} \times 1000$$

$$\text{Molarity of 2}^{\text{nd}} \text{ solution} = \frac{0.1}{\frac{0.1M + 0.9 \times 18}{d_2}} \times 1000$$

$$\Rightarrow d_2 = 0.8d_1$$

$$\frac{1}{2} \times \frac{0.2 \times 1000}{0.2M + 18 \times 0.8} \times d_1 = \frac{0.1 \times 0.8d_1}{0.1M + 0.9 \times 18}$$

$$\Rightarrow \frac{5}{0.2M + 18 \times 0.8} = \frac{4}{0.1M + 0.9 \times 18}$$

$$M = 78$$

8. **Ans (0.57)**
 $(X = 4, Y = 7)$
 $X \div Y = 0.57$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B)**
 Each ring has 15 positions total attempt = 15^3
 but only are correct attempt.
 $N = 15^3 - 1 = 2 \times 7 \times 241$ clearly divisible by 482
2. **Ans (A,B)**
 The child standing at even numbered position can not be the first child to face the problem of change.
3. **Ans (A,C)**
 $\vec{d} = x\hat{i} + y\hat{j} + z\hat{k}$ coplanar with $\vec{b}, \vec{c}$
 $\Rightarrow -3x + y - z = 0$
 projection of $\vec{d}$ on $\vec{a} = \frac{|\vec{a} \cdot \vec{d}|}{|\vec{a}|}$
 so $\sqrt{\frac{2}{3}} = \frac{1}{\sqrt{6}} |2x - y + z|$
 $2x - y + z = \pm 2$
4. **Ans (A,C)**
 $|z - z_1| + |z - z_2| = \lambda$, locus is real if $\lambda > |z_1 - z_2|$
5. **Ans (A,B,C,D)**
 $(2\lambda - 2)2 - 2(1 - 2\lambda) - (3 - \lambda) = 0 \Rightarrow \lambda = 1$
P(1, 1, -1)
 A point on L at distance r from (x_1, y_1, z_1) is
 $(x_1 + \ell r, y_1 + m r, z_1 + n r)$
 $\ell = 2/3, m = -2/3, n = -1/3, r = \pm 3$
 $Q(x = 3, y = -1, z = -2) \equiv (x_1, y_1, z_1)$
 $R(x = -1, y = 3, z = 0) \equiv (x_2, y_2, z_2)$
 Normal vector $\vec{n} = \vec{n}_1 \times \vec{n}_2 = \hat{i} + 15\hat{j} + 16\hat{k}$
 Equation of planes $1(x - 3) + 15(y + 1) + 16(z + 2) = 0$
 and $1(x + 1) + 15(y - 3) + 16(z) = 0$

6. **Ans (A,B,C,D)**

- (A) $\sqrt{x} + \sqrt{y} = 1$
 $\Rightarrow x^2 - 2xy + y^2 - 2x - 2y + 1 = \text{parabola}$
- (B) $\frac{1}{x} + \frac{1}{y} = 1$ hyperbola
- (C) $\frac{1}{x} + \frac{1}{y} + \frac{1}{x+y} = 0 \Rightarrow x^2 + 3xy + y^2 = 0$
 pair of lines
- (D) $x + y - \frac{1}{x+y} = 0 \Rightarrow x + y = \pm 1$ pair of lines

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A)**
 (A) using 2nd definition of ellipse
- (B) Let $\alpha = 3h - 2, \beta = 3k \Rightarrow \frac{\alpha + 2}{3} = h, \frac{\beta}{3} = k$
- (C) 2nd definition of hyperbola
- (D) circle with given diagonal as diameter
8. **Ans (D)**
 (A) $a_i \in P, a_i \in Q$
 (B) $a_i \in P, a_i \notin Q$
 (C) $a_i \notin P, a_i \in Q$
 (D) $a_i \notin P, a_i \notin Q$
 For $P \cup Q = A$ or $P \cap Q = \phi$, no of ways = 3^n
 singleton set in nC_1 ways. Rest of the elements in 3^{n-1} ways $P \cup Q = \phi$ in just one way.

10. Ans (B)

$$A \rightarrow R; B \rightarrow P, R; C \rightarrow Q, R, S; D \rightarrow P, Q, R$$

$$(A) \sin^2 2x + \cos^4 2x + 2 = \frac{39}{16} + \frac{1}{16} \cos 8x$$

$$(B) (2^x - 2)^2 + 1 + ||\lambda - 1| - 3| = |\sin y|$$

$$\Rightarrow \lambda - 1 = \pm 3$$

$$(C) f(x) < a \text{ for at least one real } x, x = 2, 4, 6.$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (3.00)

$$\tan \theta = t$$

$$\int e^t \frac{(1-t)^2}{(1+t^2)} dt = \frac{e^t}{1+t^2} + K, A + B = 3$$

2. Ans (5.00)

$$= \lim_{h \rightarrow \infty} \frac{1}{5^{n-1}} \sum_{\lambda=1}^n {}^n C_r \left(\sum_{i=0}^{r-1} {}^r C_i 3^i \right)$$

$$= \lim_{h \rightarrow \infty} \frac{1}{5^{n-1}} \sum_{r=1}^n {}^n C_r (4^r - 3^r)$$

$$= \lim_{n \rightarrow \infty} \frac{5^n - 4^n}{5^{n-1}} = 5$$

3. Ans (5.00)

$$z_1^3 - 3z_1 z_2^2 + 3iz_1^2 z_2 - iz_2^3 = 2 + 11i$$

$$|z_1^2 + z_2^2| = |(z_1 + iz_2)(z_1 - iz_2)| = 5$$

4. Ans (4.00)

Conceptual

6. Ans (0.00)

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{bmatrix}, B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \\ b_5 & b_6 \end{bmatrix} \Rightarrow |BA| = 0$$

7. Ans (5.00)

$$y^2 = P(x), 2y \, dydx = P'(x) \, dydx = P'(x)$$

$$\Rightarrow 2y \frac{dy}{dx^2} + 2y_1^2 = P''(x) \Rightarrow 2y^3 y_2 + 2y^2 y_1^2 = y^2 P''(x)$$

8. Ans (0.00)

$$[\sqrt{2046}] = [\sqrt{2047}] = [\sqrt{2048}] = [\sqrt{2049}] = 45$$

$$2003^{\text{rd}} \text{ term} = 2003 + 45 = 2048$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

FULL SYLLABUS

15-05-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	B	D	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,C	A,B,D	A,C	A,C	A,C,D		
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	2.00	4.00	4.00	169.40 to 169.60	20.00	3.00	6.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	D	C	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	B,C	A,B	B,C,D	A,B,C,D	A,B,C,D		
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	0.33 or 0.34	99.98	1.25	6.00	4.00	0.375	1.92 to 1.93

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,D	B,C	C,D	A,B,C	A,C,D	A,C		
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	100.00	4.00	25.00	5.20	2.00	8.00	3.00	3.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

$$\text{Resultant amplitude} = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \Delta\phi}$$

$$a^2 = 2a^2 + 2a^2 \cos(\phi_1 - \phi_2)$$

$$\Rightarrow \cos(\phi_1 - \phi_2) = -\frac{1}{2}$$

$$\phi_1 - \phi_2 = 120^\circ \text{ or } \text{rad } \frac{2\pi}{3}$$

OR

For superposition of two oscillations

$$a \sin(\omega t + \phi) = a \sin(\omega t + \phi_1) + a \sin(\omega t + \phi_2)$$

$$\Rightarrow \sin \omega t \cos \phi + \cos \omega t \sin \phi = \sin \omega t \cos \phi_1 +$$

$$\cos \omega t \sin \phi_1 + \sin \omega t \cos \phi_2 + \cos \omega t \sin \phi_2$$

$$\Rightarrow \cos \phi = \cos \phi_1 + \cos \phi_2 \text{ \& }$$

$$\sin \phi = \sin \phi_1 + \sin \phi_2$$

$$\Rightarrow \cos^2 \phi + \sin^2 \phi = 1 = \cos^2 \phi_1 + \cos^2 \phi_2 +$$

$$2\cos \phi_1 \cos \phi_2 + \sin^2 \phi_1 + \sin^2 \phi_2 + 2\sin \phi_1 \sin \phi_2$$

$$\Rightarrow \cos(\phi_1 - \phi_2) = -\frac{1}{2} \Rightarrow \phi_1 - \phi_2 = 120^\circ \text{ or }$$

$$\frac{2\pi}{3} \text{ rad}$$

2. **Ans (B)**

Field inside conductor is zero. Also field on the surface is perpendicular to the surface for conductor.

3. **Ans (D)**

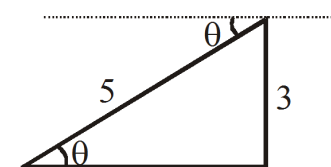
E & B are perpendicular so (4) is not possible

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,C)**

$$V_{\text{Ring1}} = V_{\text{Ring2}}$$



$$\sin \theta = \frac{3}{5}$$

$$\Rightarrow \theta = 37^\circ$$

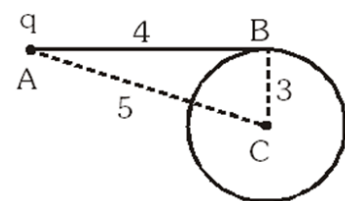
$$V_{\text{Ring1}} \times \cos 37^\circ = 5 \text{ m/sec}$$

$$\Rightarrow V_{\text{Ring1}} = \frac{5}{4} \times 5 = \frac{25}{4} \text{ m/sec}$$

6. **Ans (A,C)**

$$V = \frac{kq}{AC} = \frac{9 \times 10^9 \times 1 \times 10^{-6}}{5}$$

$$= 1.8 \times 10^3 = 1.8 \text{ kV}$$



$$V_B = (V_B)_{\text{due to } q} + (V_B)_i$$

Here $(V_B)_i$ = Potential at B due to induced charge

$$\therefore 1.8 \times 10^3 = \frac{kq}{AB} + (V_B)_i$$

$$\Rightarrow 1.8 \times 10^3 = 2.25 \times 10^3 + (V_B)_i$$

$$\Rightarrow (V_B)_i = -0.45 \text{ kV}$$

7. **Ans (A,B,D)**

$$V_A = \frac{Kq_A}{R} + \frac{Kq_B}{2R} = 2V,$$

$$V_B = \frac{K(q_A + q_B)}{2R} = \frac{3}{2}V$$

$$\Rightarrow \frac{q_A}{q_B} = \frac{1}{2}$$

After earthing of B, $q'_A = q_A$ &

$$V'_B = \frac{K(q'_A + q'_B)}{2R} = 0$$

$$\Rightarrow \left| \frac{q'_A}{q'_B} \right| = 1 \text{ \& } V'_A - V'_B = \frac{V}{2}$$

8. **Ans (A,C)**

Q value for fusion =

$$(B/A)_{75} \times 75 - [(B/A)_{30} \times 30 + (B/A)_{45} \times 45]$$

$$= 99 \text{ MeV.}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (3.00)**

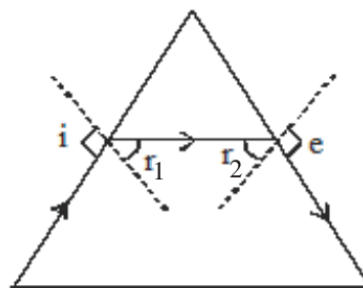
Frame will complete vertical circular motion if it crosses position where line joining O and centre of mass C makes an angle

$$\theta = \tan^{-1} \frac{a}{g} = 37^\circ \text{ with vertical}$$

By work energy theorem

$$\frac{1}{2} (2m\ell^2) \omega^2 = 2mg \frac{\sqrt{3}}{2} \ell (1 + \cos \theta) + 2ma \frac{\sqrt{3}}{2} \ell \sin \theta$$

2. **Ans (2.00)**



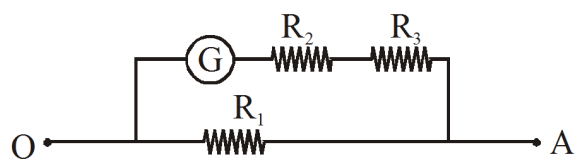
$$\sin i = \sqrt{\mu^2 - 1} \sin r_1 - \cos r_2$$

since light strike grazingly so $i = 90^\circ$ r_1 and $r_2 = 60^\circ$

$$\text{so, } \mu = 2$$

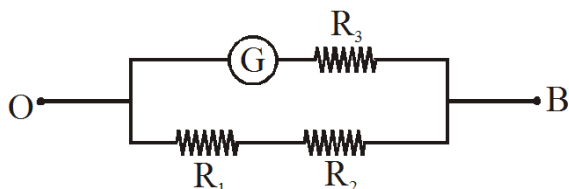
3. Ans (4.00)

For connection between OA,



$$\therefore 10 \times 10^{-3} = \frac{R_1}{R_1 + R_2 + R_3 + R_G} \times 10 \quad \dots(i)$$

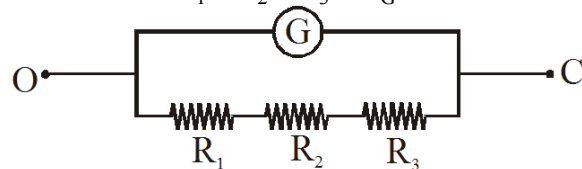
For connection between OB,



$$10 \times 10^{-3} = \frac{R_1 + R_2}{R_G + R_1 + R_2 + R_3} \times 5 \quad \dots(ii)$$

For connection between OC,

$$10 \times 10^{-3} = \frac{R_1 + R_2 + R_3}{R_1 + R_2 + R_3 + R_G} \times 1 \quad \dots(iii)$$

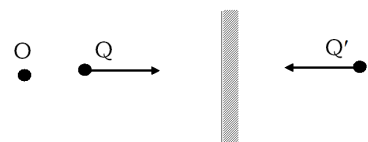


Solving (i), (ii) & (iii)

$$R_1 = \frac{1}{10} \Omega, R_2 = \frac{1}{10} \Omega \text{ \& } R_3 = \frac{4}{5} \Omega$$

5. Ans (169.40 to 169.60)

Let f_1 and f_2 be the frequencies of tuning forks P and Q, Then $|f_1 - f_2| = 7/2$



Apparent frequency for O corresponding to signal directly coming from

$$Q = f_2 \left(\frac{v}{v + v_q} \right)$$

$$\text{Apparent frequency of the echo} = f_2 \left(\frac{v}{v - v_q} \right)$$

$$\therefore \Delta f_2 = f_2 \left[\frac{2v_q v}{v^2 - v_q^2} \right]$$

Since, $\Delta f_2 = 5$ (given), $\therefore f_2 = 166$ Hz. Now, $f_1 = 166 \pm 3.5 = 169.50$ or 162.50 Hz.

When P is filed, its frequency will increase, since it is given that filed P gives greater number of beats with Q. It implies that f_1 must be 169.50 Hz.

6. Ans (20.00)

Velocity of first block before collision.

$$v_1^2 = 1^2 - 2(4) \times 0.08 = 1 - 0.64$$

$$v_1 = 0.6 \text{ m/s}$$

By conservation of momentum, $3 \times 0.6 = 3v_1' + 6v_2'$

also $v_2' - v_1' = v_1$ for elastic collision

It gives $v_2' = 0.4 \text{ m/s}$

$$v_1' = -0.2 \text{ m/s}$$

Now distance moved after collision

$$s_1 = \frac{(0.4)^2}{2 \times 4} \text{ \& } s_2 = \frac{(0.2)^2}{2 \times 4}$$

$$\therefore x = s_1 + s_2 = 0.025 \text{ m} = 2.5 \text{ cm.}$$

7. Ans (3.00)

$$\frac{1}{2} \times \left(\frac{m\ell^2}{3} \right) \left(\frac{V}{\ell} \right)^2 = mg \left(\frac{\ell}{2} \right)$$

$$\Rightarrow V = \sqrt{3g\ell} = 3$$

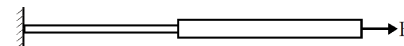
8. Ans (6.00)

$$\frac{F}{A_1} = y \frac{\Delta \ell_1}{2\ell}$$

$$\frac{F}{A_2} = y \frac{\Delta \ell_2}{\ell}$$

$$\Delta \ell_1 + \Delta \ell_2 = \frac{27}{4} \text{ mm}$$

$$\frac{2F\ell}{A_1 y} + \frac{F\ell}{4A_1 y} = \frac{27}{4} \text{ mm}$$

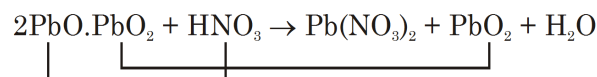


PART-2 : CHEMISTRY

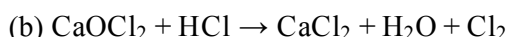
SECTION-I (i)

1. Ans (A)

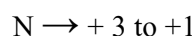
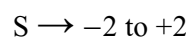
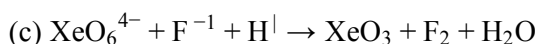
(a)



Here, PbO is behaving as base while PbO₂ is filtered out.



Cl is undergoing comproportionation



3. Ans (C)

At low pressure,

$$\left(P + \frac{a}{V^2}\right)(V) = RT$$

$$\text{i.e. } PV + \frac{a}{V} = RT$$

$$\text{i.e. } PV^2 - RTV + a = 0$$

$$V = \frac{RT \pm \sqrt{R^2T^2 - 4Pa}}{2P}$$

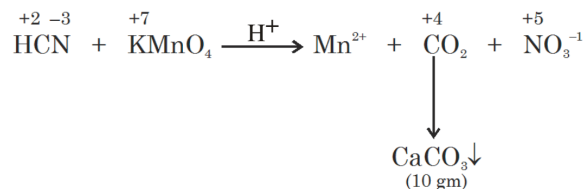
$$\text{as } R^2T^2 = 4Pa$$

$$\therefore V = \frac{RT}{2P}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (A,B)



n-factor of HCN = 10

$$\therefore N = 1 \times 10 \Rightarrow 10N$$

$$\text{Eq. mass} = \frac{M}{n\text{-factor}} = \frac{M}{10} \Rightarrow E < M$$

At eq. point :

$$N_1V_1 = N_2V_2 \Rightarrow 10 \times 100 = 5 \times 5 \times V$$

$$\Rightarrow V = 40\text{ml}$$

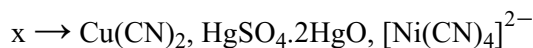
8. Ans (B,C,D)

Refer theory

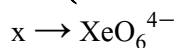
PART-2 : CHEMISTRY

SECTION-II

1. Ans (3.00)



2. Ans (0.33 or 0.34)



3. Ans (99.98)

$[\text{Ag}^+]$ for pptⁿ of I^-

$$= \frac{k_{sp}(\text{AgI})}{[\text{I}^-]} = \frac{10^{-17}}{0.1} = 10^{-16} \text{M}$$

$[\text{Ag}^+]$ for pptⁿ of Br^-

$$= \frac{k_{sp}(\text{AgBr})}{[\text{Br}^-]} = \frac{10^{-13}}{0.2} = \frac{1}{2} \times 10^{-12} \text{M}$$

$\Rightarrow [\text{I}^-]$ when AgBr starts precipitating

$$= \frac{k_{sp}(\text{AgI})}{[\text{Ag}^+]} = \frac{10^{-17}}{10^{-12}} \times \frac{1}{2} = 2 \times 10^{-5}$$

$\Rightarrow \% [\text{I}^-]$ remaining in solution

$$= 2 \times \frac{10^{-5}}{0.1} \times 100 \text{ for precipitation of AgBr}$$

$$\Rightarrow \% [\text{I}^-] \text{ precipitated} = 100 - 0.02 = 99.98$$

4. Ans (1.25)

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$T_1 = T, T_2 = \frac{T}{2}$$

$$V_1 = V \text{ and } V = 5.66 V$$

$$TV^{\gamma-1} = \frac{T}{2}(5.66V)^{\gamma-1};$$

$$2 = (5.66)g^{\gamma-1}$$

$$\gamma = 1.4$$

$$|w| = \left| \frac{P_2V_2 - P_1V_1}{\gamma - 1} \right| = \frac{nRT}{2(\gamma - 1)} = \frac{1}{0.8}PV = 1.25PV$$

5. Ans (6.00)

(i), (ii), (iv), (v), (vi), (viii)

7. Ans (0.375)

Leucine; Lysine; Threonine are essential amino acid;

$$\therefore x = 3 \text{ linkage in phosphodiester is } 3 - 5$$

$$\therefore y = 3; z = 5$$

$$\therefore \text{Ans} = \frac{3}{8} = 0.375$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**

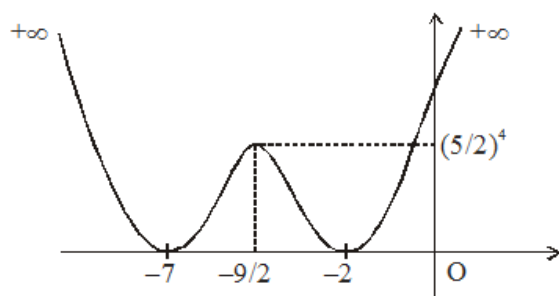
$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\ln(\sec(ex) \dots \sec(e^{50}x))}{e^{2 \cos x} \frac{(e^{2-2 \cos x} - 1)}{2-2 \cos x} \cdot \frac{2-2 \cos x}{x^2} \cdot x^2} (\%) \\ = \lim_{x \rightarrow 0} \frac{\ln(\sec(ex) \cdot \sec(e^2x) \dots \sec(e^{50}x))}{e^{2x^2}} (\%) \\ = \lim_{x \rightarrow 0} \frac{\ln(1 + \sec(ex) \dots \sec(e^{50}x) - 1)}{e^{2x^2}} \\ \left[\text{Use } \frac{\ln(1+x)}{x} \lim it \right] \\ = \lim_{x \rightarrow 0} \frac{\sec(ex) \cdot \sec(e^2x) \dots \sec(e^{50}x) - 1}{e^{2x^2}} \end{aligned}$$

Apply LH rule

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{e \sec(ex) \tan(ex) \cdot \sec(e^2x) \dots \sec(e^{50}x) + \text{so on}}{2e^{2x}} \\ &= \lim_{x \rightarrow 0} \frac{e^2 + e^4 + \dots + e^{100} \rightarrow (G.P.)}{2e^2} \\ &= \lim_{x \rightarrow 0} \frac{e^2((e^2)^{50} - 1)}{2e^2(e^2 - 1)} = \frac{e^{100} - 1}{2(e^2 - 1)} \end{aligned}$$

2. **Ans (C)**

$$\begin{aligned} \text{let } f(x) &= (x+2)^2(x+7)^2 \\ f'(x) &= 2(x+2)(x+7)^2 + (x+2)^2(x+7) \\ &= 2(x+7)(x+2)(2x+9) \end{aligned}$$



for $a = 0$, given equation has exactly two double roots ($x = -2, x = -7$).

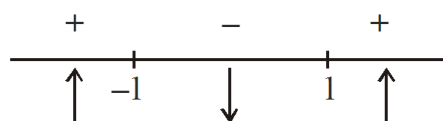
for $a = -\left(\frac{5}{2}\right)^4$, given equation has exactly one double root. $\left(x = -\frac{9}{2}\right)$

[double root $\Rightarrow$ function and the derivative both are zero at that x]

3. **Ans (B)**

$$f(x) = x^3 - 3x + 1$$

$$f'(x) = 3(x^2 - 1)$$



$$f(-1) = 3, f(0) = 1, f(2) = 3$$

Checking other nearby points.

$$f(-2) = -1, f(1) = -1$$

$$f(-2), f(-1) < 0 \quad (0, 1) \quad (1, 2)$$

$(-2, -1)$ one root

$$x_1 \in (-2, -1), x_2 \in (0, 1), x_3 \in (1, 2)$$

$$[x_1] = -2 \quad [x_2] = 0 \quad [x_3] = 1$$

$$\text{and } x_1 + x_2 + x_3 = 0 \quad (\text{sum of roots})$$

$$\therefore \{x_1\} + \{x_2\} + \{x_3\}$$

$$= x - [x_1] + x - [x_2] + x - [x_3]$$

$$= 0 - (-2 + 0 + 1) = 1$$

4. **Ans (B)**

$$P_1 + \lambda P_2 = 0$$

$$(x + 2y + z - 10) + \lambda(3x + y - z - 5) = 0$$

passing through $(0, 0, 0)$

$$\Rightarrow -10 + \lambda(-5) = 0 \Rightarrow \lambda = -2$$

$$\therefore \text{plane is } (x + 2y + z - 10) - 2(3x + y - z - 5) = 0$$

$$\Rightarrow 5x - 3z = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,D)**

$$\text{Let } \int_0^{\pi/4} \cos t \cdot f(t) dt = A$$

$$\therefore f(x) = \sin^2 x \cos^{-5} x + \mu A \sin^2 x \cos^{-5} x$$

$$= (1 + \mu A) \sin^2 x \cos^{-5} x$$

$$A = \int_0^{\pi/4} \cos t (1 + \mu A) \sin^2 t \cos^{-5} t dt$$

$$A = (1 + \mu A) \int_0^{\pi/4} \frac{\sin^2 t}{\cos^4 t} dt$$

$$= (1 + \mu A) \int_0^{\pi/4} \sec^2 t \tan^2 t dt$$

$$[\text{put } \tan t = z \sec^2 t dt = dz]$$

$$A = (1 + \mu A) \left(\frac{\tan^3 t}{3} \right)_0^{\pi/4} = \frac{1 + \mu A}{3}$$

$$\Rightarrow 1 + \mu A = 3A \text{ and } (3 - \mu)A = 1 \Rightarrow A = \frac{1}{3 - \mu}$$

$$\therefore f(x) = \frac{3}{3 - \mu} \sin^2 x \cos^{-5} x$$

$$(A)(D) \text{ period} = 2\pi$$

7. **Ans (C,D)**

$$\vec{b} = 2\hat{c} + \lambda\hat{a}$$

$$|\vec{b}|^2 = 4 + \lambda^2 + 4\lambda\hat{c} \cdot \hat{a}$$

$$|\vec{b}|^2 = 4 + \lambda^2 + 4\lambda \left(\frac{1}{7} \right) = 16$$

$$\Rightarrow \lambda^2 + \lambda - 12 = 0$$

$$\Rightarrow \lambda = -4, 3$$

8. **Ans (A,B,C)**

$$(A) |Z| = r, \arg z = \theta, z = x + iy$$

$$|x + iy - 2 - i| r \left| \sin \left(\frac{\pi}{4} - \theta \right) \right|$$

$$|(x - 2) + i(y - 1)| = r \left| \sin \frac{\pi}{4} \cos \theta - \cos \frac{\pi}{4} \sin \theta \right|$$

$$|(x - 2) + i(y - 1)| = r \left(\frac{1}{\sqrt{2}} \cos \theta - \frac{1}{\sqrt{2}} \sin \theta \right)$$

$$\sqrt{(x - 2)^2 + (y - 1)^2} = \frac{1}{\sqrt{2}} |x - y|$$

$$x^2 + 4 - 4x + y^2 + 1 - 2y = \frac{1}{2} (x - y)^2$$

$$\Rightarrow (x + y)^2 = (4x + 2y - 5)$$

$$(B) z = \text{constant} \Rightarrow \text{incorrect}$$

$$(C) z = a\omega, a\omega^2 \Rightarrow \arg = \frac{2\pi}{3}$$

9. **Ans (A,C,D)**

$$\frac{1}{y^3} \frac{dy}{dx} - \frac{1}{y^2} \cdot \frac{1}{x} = -(1 + \cos x)$$

$$\text{let } \frac{-1}{y^2} = v$$

$$\frac{2}{y^3} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{1}{2} \frac{dv}{dx} + \frac{v}{x} = -(1 + \cos x)$$

$$\Rightarrow \frac{dv}{dx} + \frac{2}{x} v = -2(1 + \cos x)$$

$$\text{u. } v^2 = -2 \int x^2 (1 + \cos x) dx$$

$$= \frac{x^3}{3} + v^2 \sin x + 2x \cos x - 2 \sin x + k$$

10. Ans (A,C)

$$I = 2 \int_0^{\pi/2} \sin x (1 - \cos^2 x) e^{-\sin^2 x} dx$$

$$2 \int_0^{\pi/2} \sin x e^{-\sin^2 x} dx + \int_0^{\pi/2} \underbrace{(\cos x)}_I \underbrace{(-\sin 2x)}_{II} e^{-\sin^2 x} dx$$

(IBP)

$$2 \int_0^{\pi/2} \sin x e^{-\sin^2 x} dx + \left(\cos x \cdot e^{-\sin^2 x} \right)_0^{\pi/2} + \int_0^{\pi/2} \sin x \cdot e^{-\sin^2 x} dx$$

$$= 3 \int_0^{\pi/2} \sin x \cdot e^{-\sin^2 x} dx - 1$$

put $-\sin^2 x = t \Rightarrow \cos^2 x = 1 + t$

$-2 \sin x \cos x dx = dt$

$$\sin x dx = \frac{-dt}{2 \cos x} = \frac{-dt}{2\sqrt{1+t}}$$

$$I = \frac{-3}{2} \int_0^{-1} \frac{e^t dt}{\sqrt{1+t}} - 1 \quad (\text{put } 1+t=z)$$

$$= \frac{-3}{2} \int_1^0 \frac{e^{z-1}}{\sqrt{z}} dz$$

$$= \frac{3}{2e} \int_0^1 e^z \frac{1}{\sqrt{z}} dz - 1 \quad (\text{IBP})$$

$$= \frac{3}{2e} \left([2\sqrt{z} e^z]_0^1 - 2 \int_0^1 e^z \sqrt{z} dz \right) - 1$$

$$= 2e$$

$$= 3 - \frac{3}{e} \int_0^1 e^z \sqrt{z} dz - 1$$

$$= (2) - (3) e \int_0^1 e^2 \sqrt{z} dz \Rightarrow \alpha = 2, \beta = 3$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (100.00)

$$[\{x\}] = 0$$

$$f(x) = (3 - x^7)^{\frac{1}{7}}$$

2. Ans (4.00)

$${}^9C_3 \times {}^3C_2 \times \frac{5!}{2!2!1!}$$

3. Ans (25.00)

$$(\sin x + \cos x)(\tan^2 x - (1 + \sqrt{3}) \tan x + 3 + \sqrt{3}) = 3 |\sin x + \cos x|$$

Case I :

$$\sin x + \cos x \geq 0$$

$$(s + c)(t^2 - (1 + \sqrt{3})t + \sqrt{3}) = 0$$

$$\Rightarrow t = -1, 1, \sqrt{3}$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{7\pi}{4}$$

Case II :

$$s + c < 0$$

$$(s + c)(t^2 - (1 + \sqrt{3})t + 6 + \sqrt{3}) = 0$$

$$\Rightarrow t = -1 \quad \text{No solution}$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\Rightarrow \text{sum of solutions is } \frac{37\pi}{12}$$

$$|p - q| = 25$$

4. Ans (5.20)

$$t_1 = \frac{\sin x}{\cos 3x}, t_2 = \frac{\sin 3x}{\cos 9x}, t_3 = \frac{\sin 9x}{\cos 27x}$$

$$t_1 = \frac{\sin x \cdot 2 \cos x}{\cos 3x \cdot 2 \cos x} = \frac{\sin 2x}{2 \cos x \cos 3x} = \frac{\sin(3x - x)}{2 \cos x \cos 3x}$$

$$= \frac{1}{2} (\tan 3x - \tan x)$$

$$\text{Similarly, } t_2 = \frac{1}{2} (\tan 9x - \tan 3x)$$

$$t_3 = \frac{1}{2} (\tan 27x - \tan 9x)$$

$$t_1 + t_2 + t_3 = 0 \Rightarrow \frac{1}{2} (\tan 27x - \tan x) = 0$$

$$\Rightarrow \tan 27x = \tan x$$

$$\Rightarrow 27x = n\pi + x$$

$$\Rightarrow 26x = n\pi \Rightarrow x = \frac{n\pi}{26}$$

$$n = 1 \Rightarrow x = \frac{\pi}{26} \therefore k = 26$$

5. Ans (2.00)

$$x + y + z = \theta \Rightarrow \cos x + \cos y + \cos z = 2\cos\theta$$

$$\& \sin x + \sin y + \sin z = 2\sin\theta$$

$$\& \cos(x + y) + \cos(y + z) + \cos(z + x) = ?$$

$$= \cos(\theta - z) + \cos(\theta - x) + \cos(\theta - y)$$

$$= \cos\theta \cos z + \sin\theta \sin z + \cos\theta \cos x + \sin\theta \sin x + \cos\theta \cos y + \sin\theta \sin y$$

$$= \cos\theta(\cos x + \cos y + \cos z) + \sin\theta(\sin x + \sin y + \sin z)$$

$$= 2\cos^2\theta + 2\sin^2\theta = 2$$

6. Ans (8.00)

$$\text{Replace } x \rightarrow \frac{2}{x}$$

$$\left(\frac{8}{x^2} + \frac{6}{x} + 4\right)^{10} = \sum_{r=0}^{20} a_r \left(\frac{2}{x}\right)^r$$

$$\Rightarrow \frac{2^{10}(2x^2 + 3x + 4)^{10}}{x^{20}} = \sum_{r=0}^{20} a_r \cdot 2^r \cdot x^{-r}$$

$$\Rightarrow 2^{10}(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r \cdot 2^r \cdot x^{20-r}$$

$$\Rightarrow 2^{10} \sum_{r=0}^{20} a_r \cdot x^r = \sum_{r=0}^{20} a_r \cdot 2^r \cdot x^{20-r}$$

$$\therefore \text{coefficient of } x^7$$

$$2^{10} a_7 = a_{13} 2^{13}$$

$$\Rightarrow \frac{a_7}{a_{13}} = 2^3 = 8$$

7. Ans (3.00)

$$\text{Let } f(x) = \tan^{-1}x, x \in [a, b]$$

$$\therefore \text{Using LMVT, we get } \frac{\tan^{-1}b - \tan^{-1}a}{b - a} = \frac{1}{1 + c^2},$$

$$\text{where } 0 < a < c < b < \sqrt{3}$$

$$\text{So, } 1 < \left(\frac{b - a}{\tan^{-1}b - \tan^{-1}a} \right) < 4$$

$$\text{As, } \frac{1}{4} < \frac{1}{1 + c^2} < 1$$

$$\Rightarrow \text{The greatest possible integral value is 3.}$$

8. Ans (3.00)

$$\int_0^{\pi/2} \frac{dx}{\cos^2 x (1 + \sqrt{\tan x})^4} = \int_0^{\pi/2} \frac{\sec^2 x}{(1 + \sqrt{\tan x})^4} dx$$

$$\text{Put } \tan x = t^2 \Rightarrow \sec^2 x dx = 2t dt$$

$$\int_0^{\infty} \frac{2t}{(1+t)^4} dt = 2 \int_0^{\infty} \left(\frac{1}{(1+t)^3} - \frac{1}{(1+t)^4} \right) dt$$

$$= 2 \left(\frac{-1}{2(1+t)^2} + \frac{1}{3(1+t)^3} \right)_0^{\infty}$$

$$= \left(\frac{-1}{2(1+t)^2} + \frac{1}{3(1+t)^3} \right)_0^{\infty}$$

$$= 2 \left(0 - \left(\frac{-1}{2} + \frac{1}{3} \right) \right)$$

$$= 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}$$